

Thematic Alignment Analysis

Journal of Biomedical Informatics (2015–2024)

Analyzing Thematic Alignment in Journal of Biomedical Informatics

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Overview

01

Motivation

To what extent have articles published in JBI drifted from the journal's stated Aims & Scope over the period 2015–2024, and how has this alignment changed over time?

02

Data

2,039 abstracts from JBI via PubMed API
2015–2024 (10 years)

03

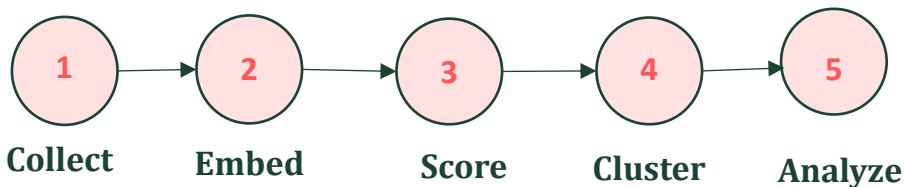
Method

Sentence-BERT embeddings + Cosine Similarity
K-Means clustering + UMAP visualization

04

Finding

Measurable thematic drift detected
AI/ML articles show lower alignment scores



Dataset

2,039

Abstracts

Why this data?

JBIs sits at the intersection of medicine and computer science , making it uniquely sensitive to the AI/ML wave

244

Mean abstract length (words)

10

Years (2015–2024)

28

Outliers detected

Source: PubMed API • Query: "J Biomed Inform"[Journal] AND 2015–2024 • Excluded: no abstract, 2025 indexing artifacts

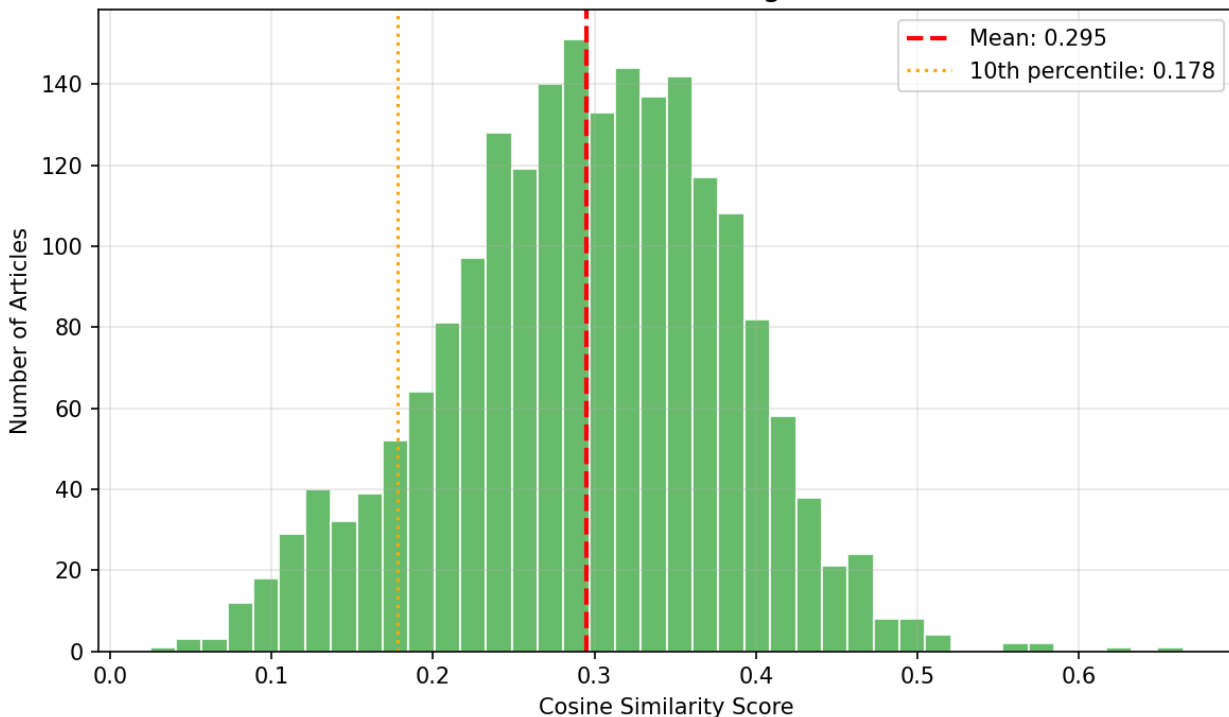
Embedding Model Selection

	all-MiniLM-L6-v2 ✓	S-PubMedBERT
Score range	0.025 – 0.665	0.845 – 0.946
Mean score	0.295	0.898
Std dev	0.09	0.02
Optimal k	2	7
Drift detection	✓ Suitable	✗ Too narrow
Cluster quality	Generic (2)	Granular (7)

all-MiniLM-L6-v2 was selected: wider score range gives better discriminative power for drift detection.

Alignment Scores

Distribution of Thematic Alignment Scores



Normal-curve distribution — low mean expected (short Aims & Scope vs. detailed abstracts)

0.295

Mean score

0.025

Min score

0.665

Max score

1.4%

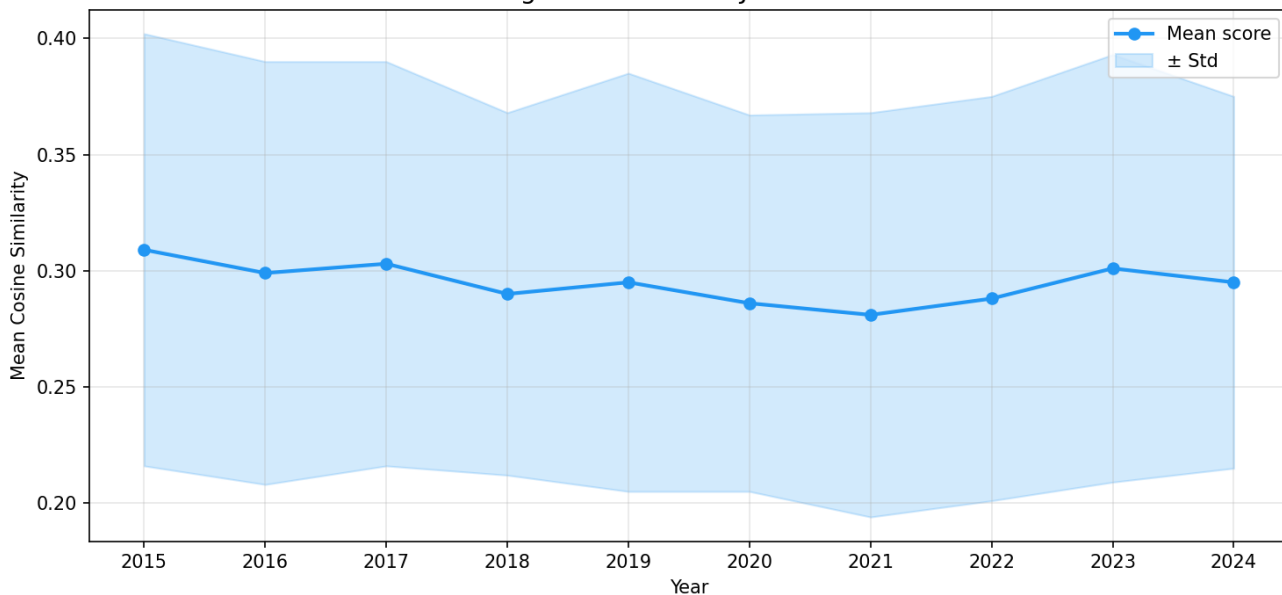
Outliers (< 0.1)

Why cosine similarity?

- Measures semantic direction, not magnitude
- Abstract length does not affect the score

Thematic Drift Over Time

Thematic Alignment Score of JBI Articles Over Time



0.309

2015 (peak)

0.281

2021 (lowest)

0.301

2023 (recovery)

5.8%

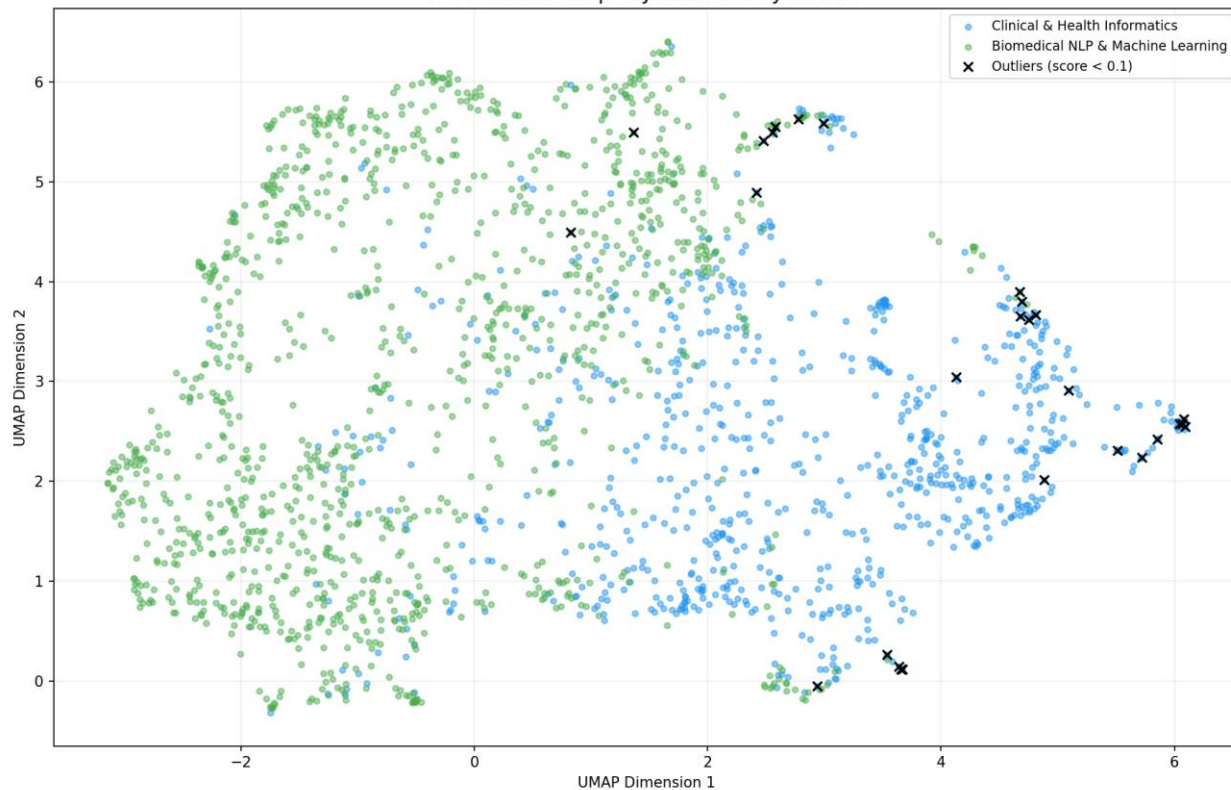
Highly aligned in 2018

- 2015–2018: gradual decline as ML topics enter the field
- 2018–2021: steepest drop; COVID-19 + AI/ML surge
- 2022–2024: partial recovery as editorial focus stabilizes

*Gradual decline 2015→2021,
partial recovery 2022–2023*

Cluster Analysis

2D Semantic Map of JBI Articles by Cluster



Cluster 0

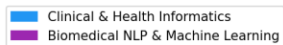
Clinical & Health Informatics
health, patient, research

Cluster 1

Biomedical NLP & ML
learning, disease, medical

- Outliers cluster on the right edge
 - ✓ Right side = furthest from the journal's core topics
- These articles are semantically distant from both clusters
- Confirms they are genuinely off-topic, not just borderline cases

k=2 (Silhouette Score) • Low separation expected — JBI is a focused journal

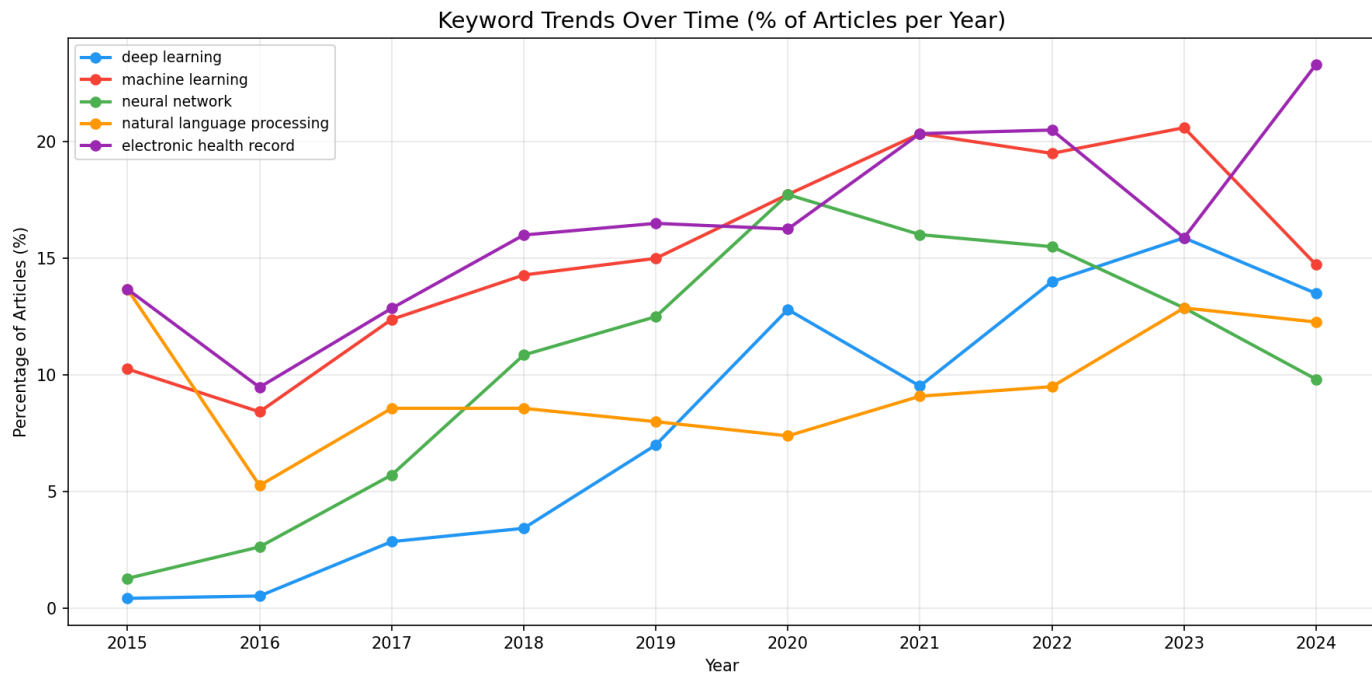


Thematic Composition by Year



Clinical & Health Informatics: 42% → 26% Biomedical NLP & ML: 58% → 74% (2015 → 2024)

Keyword Trends



deep learning

0.4% → 16%

machine learning

10% → 21%

neural network

1% → 16%

Rise in AI/ML terminology directly correlates with the observed thematic drift

Conclusions



Thematic drift is real and measurable in JBI (2015–2024)



Mean alignment declined from 0.309 (2015) to 0.281 (2021), then partially recovered



Biomedical NLP & ML cluster grew from 58% to 74% — and shows lower alignment



COVID-19 period caused a temporary spike in outlier articles



Drift is moderate, not severe — JBI has expanded, not departed from its scope

Limitations & Future Work

⚠ Limitations

Abstracts only

Full texts would give richer signal

Manual cluster labels

Based on TF-IDF keywords — some subjectivity

$k=2$

PubMedBERT gives $k=7$ but poor scoring range

→ Future Work

Full text analysis

Complete articles instead of abstracts

BERTopic

Automated topic labeling, less subjectivity

Multi-journal

Compare drift across several journals

Thank you for your time!